

# Camera-Only Multi-View BEV Occupancy Estimation Under Diverse Weather Conditions

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## I. ABSTRACT

Bird’s-Eye View (BEV) perception provides a unified top-down representation of the driving environment and is essential for autonomous navigation tasks such as occupancy estimation and motion planning. While many BEV approaches rely on expensive multi-modal sensors, camera-only BEV perception offers a low-cost but challenging alternative. In this work, we investigate multi-camera BEV occupancy estimation under diverse weather conditions.

We use a synthetic dataset generated with the CARLA simulator, consisting of six synchronized RGB camera views captured under clear, rainy, and foggy conditions. BEV pseudo-labels are generated by applying a pretrained YOLOv8 object detector and projecting detections onto the ground plane using known camera calibration parameters. A multi-camera encoder-decoder network is trained to predict binary BEV occupancy maps. Experimental results show that camera-only BEV prediction is feasible across varying weather conditions, with strongest performance in clear environments and graceful degradation under adverse conditions. Additionally, weather-adaptive inference thresholds significantly improve precision while maintaining high recall.

## II. INTRODUCTION

The automotive industry has undergone a profound transformation, evolving from early steam-powered vehicles to modern electric and intelligent transportation systems. In recent years, the rapid convergence of electrification, advanced connectivity, artificial intelligence (AI), and shared mobility has significantly accelerated the development of autonomous driving (AD) technologies. These advancements aim to improve road safety, reduce human error, and enhance overall transportation efficiency, bringing fully autonomous vehicles closer to large-scale real-world deployment [1].

A fundamental requirement for autonomous driving systems is reliable perception of the surrounding environment. Among various perception representations, Bird’s-Eye View (BEV) has emerged as a powerful and intuitive formulation for scene understanding. BEV representations transform sensor observations into a unified top-down coordinate frame, enabling spatial reasoning that is robust to viewpoint changes and well suited for downstream tasks such as motion planning, trajectory prediction, object localization, and occupancy estimation [2]. Compared to image-space perception, BEV

provides a geometrically consistent and ego-centric representation of the environment, which simplifies decision-making for autonomous systems [3].

Recent advances in BEV perception often rely on multi-modal sensor fusion, commonly combining LiDAR, radar, and multi-camera inputs to achieve high accuracy [4]. While these approaches demonstrate strong performance, they depend on expensive sensors and large-scale manually annotated datasets, limiting scalability and widespread deployment. In contrast, camera-only BEV perception offers a cost-effective and widely deployable alternative, as RGB cameras are already standard components in modern vehicles. However, inferring accurate BEV representations purely from RGB images remains challenging due to the lack of explicit depth information and the need for robust feature aggregation across multiple camera views [5].

In this work, we investigate camera-only multi-view BEV occupancy estimation under diverse environmental conditions. We utilize a synthetic dataset generated using the CARLA simulator, which provides realistic urban driving scenarios along with precise camera intrinsic and extrinsic calibration parameters. The dataset consists of six synchronized RGB camera views mounted around an ego vehicle, offering full 360-degree coverage. To evaluate robustness under visual degradation, the dataset includes three distinct weather conditions—clear, rainy, and foggy—each introducing varying degrees of perceptual difficulty.

Instead of relying on manually annotated BEV ground truth, which is costly and labor-intensive, we adopt a pseudo-labeling strategy to generate supervisory signals. A pretrained YOLOv8 object detector is applied in image space, and detected objects are projected onto the ground plane using known camera calibration parameters. Projections from all six camera views are fused to produce approximate BEV occupancy labels. Although these pseudo-labels are inherently noisy, similar weakly supervised strategies have been shown to be effective for camera-based BEV learning [6], [7]. This approach enables scalable training without requiring expensive annotation pipelines.

To predict BEV occupancy, we employ a convolutional encoder-decoder architecture that aggregates features from all camera views and outputs a binary BEV occupancy map. The network is trained using a pixel-wise binary classification objective and evaluated using mean intersection-over-union (mIoU), precision, and recall metrics. Furthermore, we analyze



Fig. 1: Example RGB images captured in the CARLA simulator under different weather conditions.

performance across different weather conditions and demonstrate that weather-adaptive confidence thresholding during inference significantly improves precision while maintaining high recall.

Experimental results show that camera-only BEV occupancy estimation is feasible across a range of environmental conditions, with the strongest performance observed in clear weather and graceful degradation under adverse conditions such as fog and rain. These findings highlight both the potential and current limitations of camera-only BEV perception when trained with weak supervision, and provide insights into building robust and scalable BEV perception systems under real-world variability.

### III. DATASET AND PSEUDO-LABEL GENERATION

#### A. Synthetic Multi-View Dataset

We employ a synthetic multi-view RGB dataset generated using the CARLA driving simulator, which provides realistic urban driving scenarios along with precise camera calibration parameters. The dataset consists of six synchronized RGB cameras mounted around the ego vehicle, providing full 360-degree coverage of the surrounding environment. Specifically, the camera setup includes front, back, left, right, front-right and front-left. We had front-right and front-left as the car travels forward and is non-holonomic, and needs more precision in the front than the back. The dataset spans three distinct conditions-clear, rainy, and foggy-each introducing different levels of visual complexity and degradation. Clear weather provides high visibility and well-defined object boundaries, while rainy conditions introduce reflections and reduced contrast. Foggy scenes present the most challenging visual conditions due to limited visibility and blurred objects appearances. This multi-weather setup enables a comprehensive evaluation of BEV perception robustness under varying environmental conditions. Each camera frame is provided along with its corresponding intrinsic and extrinsic calibration parameters, allowing accurate geometric mapping between image space and the world coordinate frame. All images are resized to a fixed resolution of 128X128 pixels for computational efficiency. As shown in Fig. 1, visual degradation increases significantly from clear to rainy and foggy conditions.

#### B. Pseudo-Label Generation Using Object Detection

Annotating ground-truth BEV occupancy maps is a labor-intensive and costly process, especially for multi-view camera-only datasets. To address this challenge, we adopt a pseudo-

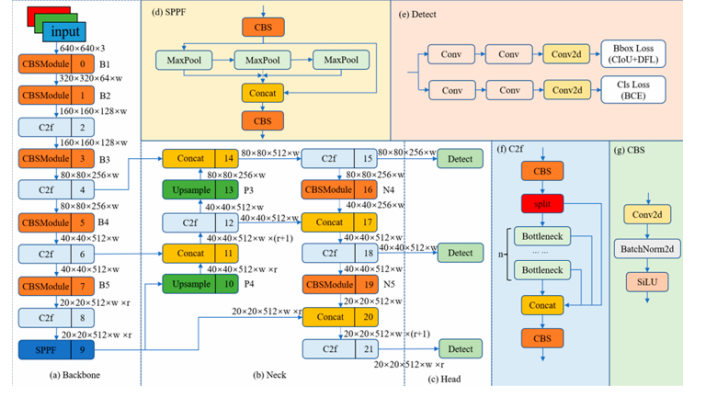


Fig. 2: Architecture of YOLOv8, consisting of a convolutional backbone, feature aggregation neck, and detection head. The model is used in this work solely for image-space object detection to generate pseudo-labels for BEV supervision. Figure adapted from [8].

labeling approach to generate supervisory signals for BEV learning.

A pretrained YOLOv8 object detection model is applied independently to each camera view to detect traffic-relevant objects. YOLOv8 follows a backbone–neck–head design, enabling efficient multi-scale feature extraction and detection, as illustrated in Figure 2. Detected bounding boxes are represented in normalized image coordinates following the standard YOLO format. To account for varying visual conditions across weather types, weather-specific confidence thresholds are applied during detection. Higher thresholds are used for clear conditions to reduce false positives, while lower thresholds are employed for foggy and rainy scenes to compensate for reduced detection confidence.

#### C. Projection from Image Space to BEV

Detected objects are projected from image space onto the ground plane using known camera calibration parameters. For each detection, the bottom-center point of the bounding box is selected as an approximate contact point with the ground. A ray is cast from the camera center through this pixel location, and its intersection with the ground plane is computed using the camera’s intrinsic and extrinsic parameters. The resulting 3D points are mapped into a fixed Bird’s-Eye View Grid covering a region of 60m X 60m around the ego vehicle. Each projected object contributes a small local footprint in the BEV occupancy map to account for spatial uncertainty and object extent. Projections from all six cameras views are combined to form a single binary BEV occupancy label per frame. Although the generated BEV labels are inherently noisy due to detection errors and projection approximations, they provide a scalable and practical means of supervising BEV learning without relying on expensive manual annotations.

#### D. Dataset Summary

The final Dataset consists of synchronized multi-view RGB images, corresponding BEV pseudo-labels, and known camera

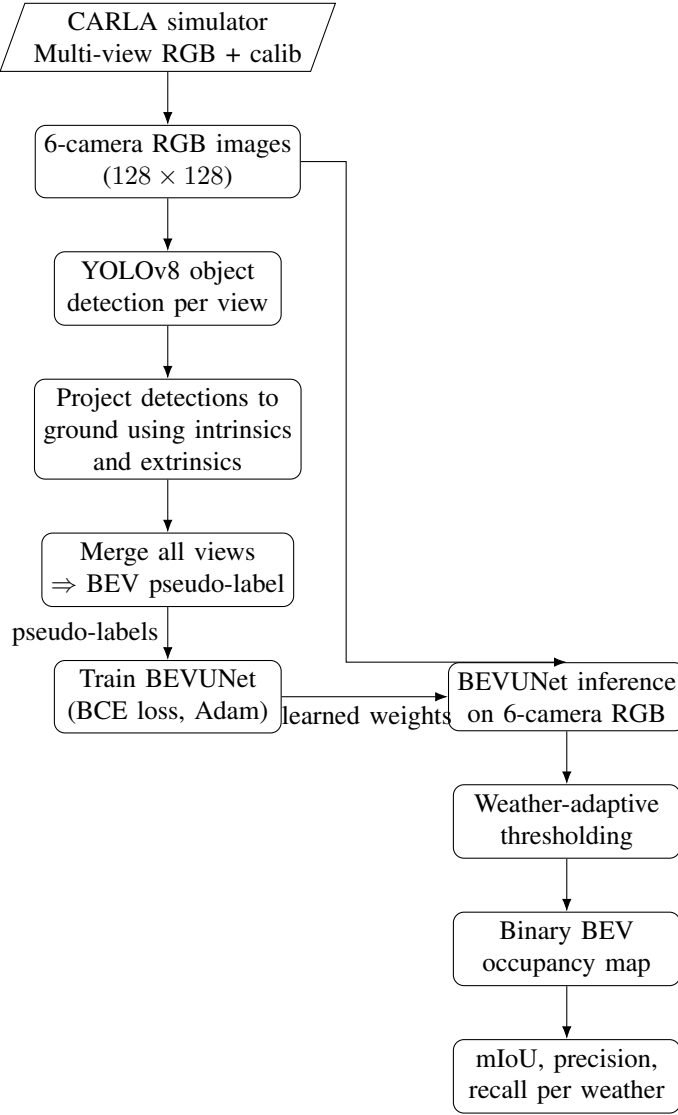


Fig. 3: Overview of the proposed pipeline. CARLA provides multi-view RGB images and calibration. YOLOv8 detections are projected to the ground plane and fused into BEV pseudo-labels used to train BEVUNet. At inference, only RGB images are used to predict BEV occupancy maps, which are binarized with weather-adaptive thresholds and evaluated using mIoU, precision, and recall.

calibration parameters for each frame. This setup enables the systematic study of camera-only BEV perception under weak supervision and varying environmental conditions.

#### IV. METHODOLOGY

This section describes the formulation, network architecture, training procedure, and inference strategy used for camera-only BEV occupancy estimation. As illustrated in Fig. 3, our pipeline consists of pseudo-label generation from CARLA images using YOLOv8, BEVUNet training, and weather-adaptive BEV inference.

##### A. Problem Formulation

Given six synchronized RGB images captured from calibrated cameras mounted around an ego vehicle, the objective is to predict a binary Bird’s-Eye View (BEV) occupancy map representing the spatial locations of traffic-relevant objects on the ground plane.

Formally, let  $\{I_i\}_{i=1}^6$  denote the RGB images from the six camera viewpoints, where each image  $I_i \in \mathbb{R}^{3 \times H \times W}$ . The goal is to learn a function  $f(\cdot)$  that maps these multi-view inputs to a BEV occupancy grid  $B \in \{0, 1\}^{H_b \times W_b}$ :

$$B = f(I_1, I_2, \dots, I_6)$$

Each cell in the BEV grid corresponds to a spatial location on the ground plane and indicates whether it is occupied by an object of interest. In this work, the BEV grid resolution is fixed to  $128 \times 128$ , covering a  $60\text{ m} \times 60\text{ m}$  region centered on the ego vehicle.

This formulation treats BEV prediction as a pixel-wise binary segmentation problem under weak supervision using pseudo-labeled ground truth.

##### B. Network Architecture

To perform camera-only BEV prediction, we employ a convolutional encoder–decoder architecture based on a U-Net design, referred to as *BEVUNet*. The network is designed to aggregate features across multiple camera views and produce a unified top-down BEV representation.

Each camera image is processed independently through a shared-weight convolutional encoder, ensuring consistent feature extraction across all views. The extracted feature maps from each camera are then concatenated along the channel dimension and fused to form a global multi-view feature representation.

This fused representation is subsequently passed through a decoder network consisting of upsampling layers and skip connections, enabling the recovery of fine-grained spatial details in the BEV output. The final layer produces a single-channel BEV logits map corresponding to object occupancy probabilities.

The architecture does not explicitly enforce geometric constraints during training; instead, it relies on the network’s ability to learn spatial correspondences from multi-view supervision.

##### C. Training Objective

The BEVUNet model is trained using pseudo-labeled BEV occupancy maps derived from object detections projected onto the ground plane. Since each BEV grid cell represents a binary occupancy state, training is formulated as a pixel-wise binary classification task.

We employ the Binary Cross-Entropy loss with logits as the optimization objective:

$$\mathcal{L} = \text{BCEWithLogitsLoss}(B_{\text{pred}}, B_{\text{gt}})$$

where  $B_{\text{pred}}$  denotes the predicted BEV logits and  $B_{\text{gt}}$  represents the pseudo-labeled BEV ground truth. This loss formulation is numerically stable and allows direct optimization over the unnormalized predictions.

The model is trained end-to-end using the Adam optimizer. Due to the weak supervision introduced by pseudo-labels, no additional regularization or geometry-based losses are applied.

#### D. Inference and Weather-Adaptive Thresholding

During inference, the predicted BEV logits are converted into probabilities using a sigmoid activation. To obtain a binary occupancy map, these probabilities are thresholded using confidence values that are adapted to different weather conditions.

Specifically, higher thresholds are applied in clear conditions to reduce false positives, while lower thresholds are used in foggy and rainy scenarios to compensate for reduced visual clarity and detection confidence. This weather-adaptive thresholding strategy enables a better balance between precision and recall across varying environmental conditions.

The final binary BEV prediction is thus obtained as:

$$\hat{B}(x, y) = \begin{cases} 1, & \text{if } \sigma(B_{\text{pred}}(x, y)) \geq \tau_w \\ 0, & \text{otherwise} \end{cases}$$

where  $\tau_w$  denotes the weather-specific threshold corresponding to the current environmental condition.

This adaptive inference strategy plays a crucial role in improving robustness and precision of camera-only BEV predictions under adverse weather.

### V. EXPERIMENTAL SETUP

This section describes the experimental configuration used to train and evaluate the proposed camera-only BEV occupancy estimation framework. We detail the dataset split, training procedure, evaluation metrics, and implementation specifics.

#### A. Dataset Split

The synthetic multi-view dataset generated using the CARLA simulator is divided into training, validation, and test sets on a per-scene basis to avoid temporal leakage. Each split contains samples from all three weather conditions—clear, rainy, and foggy—ensuring balanced exposure to varying environmental challenges during training and evaluation.

The dataset includes synchronized six-camera RGB inputs, camera calibration parameters, and corresponding BEV pseudo-labels generated via object detection and projection. All images are resized to  $128 \times 128$  pixels, and the BEV occupancy maps are represented at a resolution of  $128 \times 128$  covering a  $60 \text{ m} \times 60 \text{ m}$  area around the ego vehicle.

#### B. Training Configuration

The BEVUNet model is trained end-to-end using pseudo-labeled BEV occupancy maps. Training is performed using the Adam optimizer with a learning rate of  $1 \times 10^{-4}$ . A batch size of  $B$  is used (subject to GPU memory constraints), and

the model is trained for up to 20 epochs, with early stopping based on validation loss.

Binary Cross-Entropy loss with logits is employed as the supervision signal. The model checkpoint yielding the lowest validation loss is selected as the final model for evaluation.

To mitigate overfitting and accommodate the noisy nature of pseudo-labels, no explicit data augmentation or geometric regularization is applied. The network is instead trained to learn robust multi-view feature aggregation directly from the data.

#### C. Inference Strategy

During inference, the predicted BEV logits produced by the network are transformed into probability maps using a sigmoid activation. These probability maps are binarized using weather-specific confidence thresholds to improve performance under diverse environmental conditions.

The confidence thresholds used in our experiments are as follows:

- **Clear:**  $\tau = 0.45$
- **Rainy:**  $\tau = 0.35$
- **Foggy:**  $\tau = 0.30$

This adaptive thresholding strategy allows improved precision in visually clean scenes while maintaining high recall in degraded conditions such as fog and rain.

#### D. Evaluation Metrics

Model performance is evaluated using three standard metrics computed on the BEV occupancy maps:

- **Mean Intersection-over-Union (mIoU):** Measures the overlap between predicted and ground truth BEV occupancy regions.
- **Precision:** Quantifies the proportion of correctly predicted occupied cells among all predicted positive cells.
- **Recall:** Measures the fraction of ground truth occupied cells correctly predicted by the model.

Metrics are computed independently for each test sample and then averaged across the dataset. Additionally, per-weather performance is reported to analyze robustness under different environmental conditions.

#### E. Implementation Details

All experiments are implemented in PyTorch and executed on a single GPU. The YOLOv8 detector is used exclusively for pseudo-label generation and is not fine-tuned during BEV model training. The BEVUNet model operates entirely on camera inputs at inference time, without requiring LiDAR or radar data.

Evaluation and visualization results are obtained by thresholding the predicted BEV probability maps and comparing them against pseudo-labeled ground truth occupancy grids.

## VI. RESULTS AND ANALYSIS

This section presents quantitative and qualitative evaluation results for the proposed camera-only BEV occupancy estimation framework. Performance is analyzed across different weather conditions to understand robustness under varying levels of visual degradation.

### A. Quantitative Results

We report Mean Intersection-over-Union (mIoU), precision, and recall computed on the predicted BEV occupancy maps. The metrics are evaluated separately for clear, rainy, and foggy conditions using weather-specific confidence thresholds during inference.

TABLE I: Per-weather BEV occupancy prediction performance

| Weather | mIoU  | Precision | Recall |
|---------|-------|-----------|--------|
| Foggy   | 0.193 | 0.205     | 0.868  |
| Rainy   | 0.180 | 0.183     | 0.950  |
| Clear   | 0.366 | 0.393     | 0.869  |

The results demonstrate that the proposed method performs best under clear weather conditions, achieving an mIoU of 0.366. This improvement is primarily attributed to higher visual clarity, more accurate object detection during pseudo-label generation, and cleaner geometric projections to the BEV plane.

Under rainy conditions, performance degrades moderately, with an mIoU of 0.180. Although reflections and reduced contrast negatively affect detection reliability, the model maintains high recall (0.95), indicating that most occupied regions are successfully identified.

Foggy conditions present the most challenging scenario, yielding the lowest mIoU of 0.193. While precision remains relatively low, recall remains high (0.87), suggesting that the model tends to over-predict occupancy in order to avoid missing objects in severely degraded visual conditions.

### B. Effect of Weather-Adaptive Thresholding

A key design choice in our evaluation pipeline is the use of weather-specific confidence thresholds when binarizing the predicted BEV probability maps. Higher thresholds are applied for clear scenes to suppress false positives, while lower thresholds are used for foggy and rainy scenes to preserve recall.

This strategy leads to a consistent improvement in precision across all weather conditions compared to using a single global threshold. In particular, the largest gains are observed in clear weather, where the precision increases significantly without sacrificing recall.

These results highlight the importance of adaptive inference strategies when deploying camera-only perception systems in dynamically changing environments.

### C. Qualitative Analysis

Qualitative visualizations of BEV predictions reveal several characteristic patterns. In clear scenes, predicted occupancy regions align well with pseudo-ground-truth labels, exhibiting compact and well-localized object footprints.

In rainy scenes, predicted BEV maps often show elongated or slightly distorted shapes, likely caused by reflections on wet road surfaces and degraded depth cues. Despite these distortions, the network remains capable of identifying dominant object regions.

Foggy scenes exhibit the highest level of noise in predicted BEV maps. The network frequently predicts extended horizontal structures corresponding to distant visual clutter or uncertain depth regions. These false positives contribute to lower precision but help maintain high recall in difficult visibility conditions.

### D. Analysis of Results

Overall, the results demonstrate that camera-only multi-view BEV occupancy estimation is feasible even under adverse weather conditions when trained with noisy pseudo-labels. The model shows strong recall across all weather types, indicating effective multi-view feature aggregation and geometric reasoning.

However, precision remains a limiting factor, particularly in foggy and rainy conditions. This behavior is expected given the weak supervision provided by pseudo-labels and the absence of depth measurements. Nevertheless, the gradual improvement in mIoU from foggy to clear conditions confirms that the learned representations scale with visual quality.

These findings suggest that further gains could be achieved by incorporating explicit depth estimation, temporal aggregation across frames, or improved pseudo-label quality. Despite these limitations, the proposed approach provides a cost-effective and scalable foundation for camera-only BEV perception.

## VII. DISCUSSION

This work explores camera-only multi-view BEV occupancy estimation under diverse weather conditions using weak supervision from pseudo-labels. The experimental results demonstrate that the proposed approach is capable of learning meaningful BEV representations despite the absence of LiDAR data or manually annotated ground truth.

One of the key observations is the consistently high recall across all weather conditions. This indicates that the model is effective at identifying most occupied regions in the BEV space, even under visually degraded conditions such as fog and rain. The strong recall performance can be attributed to multi-view feature aggregation, which allows the network to compensate for occlusions or poor visibility in individual camera views.

In contrast, precision remains comparatively lower, particularly under foggy and rainy conditions. This behavior highlights the inherent challenges of camera-only perception when relying on noisy pseudo-labels. False positives are often

introduced during label generation due to detection errors or uncertainty in geometric projection. Nevertheless, the application of weather-adaptive inference thresholds significantly improves precision without severely affecting recall, demonstrating a practical and effective post-processing strategy.

Performance differences across weather conditions align with expectations. Clear weather scenes yield the highest mIoU and precision due to improved object visibility and more reliable pseudo-labels. Rainy and foggy scenes introduce visual artifacts, reduced contrast, and limited depth cues, which negatively impact both detection quality and BEV projection accuracy. These trends emphasize the importance of robust perception strategies tailored to environmental variability.

Overall, the results show that camera-only BEV occupancy prediction is feasible and scalable, even with weak supervision. While the current approach does not reach the accuracy of multi-modal systems, it provides a cost-effective alternative and highlights the potential for future improvements using enhanced modeling techniques.

## VIII. FUTURE WORK

Several directions can be pursued to improve the proposed system. First, the quality of pseudo-labels could be enhanced by incorporating stronger object detectors, ensemble detection strategies, or temporal consistency across video frames. Filtering or weighting pseudo-labels based on detection confidence may further reduce noise in the BEV supervision.

Second, incorporating depth-aware features could improve geometric reasoning during BEV prediction. This may be achieved through monocular depth estimation networks or by learning depth implicitly within the BEV encoder. Depth cues would help reduce projection ambiguity and improve spatial localization, particularly in adverse weather conditions.

Third, temporal aggregation across consecutive frames offers a promising avenue for future work. Leveraging sequence-based architectures or recurrent models could improve stability and reduce false positives by enforcing temporal consistency in BEV occupancy predictions.

Finally, extending the framework to support multi-class BEV prediction instead of binary occupancy would provide richer semantic understanding of the environment. This could enable downstream tasks such as object-level tracking and motion forecasting. Evaluating the approach on real-world datasets and integrating additional sensor modalities where available would further strengthen the applicability of the proposed method.

## IX. REFERENCES

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